BRO Academy

Manual

Pseudo

PSEUDO CLASSES

pseudo-classes

pseudo-classes (or "fake" classes) are a way to style elements based on their state or position within the document, even though these states or positions are not explicitly represented in the HTML structure (DOM). Unlike classes, which are added manually in the HTML, pseudo-classes are used to target elements in specific states, such as when a user interacts with them or based on their position in the document.

- **User-action** pseudo-classes
- **The lang** pseudo-classes
- **The negation** pseudo-classes
- **Structural** pseudo-classes
- **User interface** pseudo-classes

Link Pseudo-classes for `<a>`

`:link` – elements that have not been visited by the user. By default, browsers typically style unvisited links in blue, but with the :link pseudo-class, you can customize this appearance to match the design of your website.

`:visited` – applies styles to links that the user has already clicked on and visited. This is useful for providing visual feedback, so users can easily distinguish between the links they have visited and those they haven't. Browsers typically style visited links in purple, but you can change this using the :visited pseudo-class to fit your design.
In practice, it is also used very rarely.

User-action pseudo-classes

pseudo-classes allow you to style elements based on the way that users interact with these elements.
Start by explaining that user-action pseudo-classes are a group of CSS pseudo-classes that allow you to apply styles based on user interactions with elements on the webpage. These interactions include actions like clicking on an element, focusing on an input field, or hovering over a button or link. These pseudo-classes are essential for creating interactive and responsive designs that react to user behavior.

:active

The :active pseudo-class applies styles to an element while it is being clicked or activated. This means that the styles defined under :active are visible only during the short moment when the user is pressing down on the element.
It's commonly used to give feedback to users when they click on buttons or links, often by changing the element's appearance to indicate that it is being clicked.

:focus

The :focus pseudo-class targets elements that are currently in focus. This typically applies to form elements like input fields, text areas, or buttons when they are selected by the user, either by clicking on them or navigating to them using the keyboard (e.g., using the Tab key).
Styling elements with :focus is crucial for accessibility, ensuring that users who navigate your site via keyboard or assistive devices can easily see which element is currently active.

:hover

The :hover pseudo-class is triggered when the user hovers the mouse pointer over an element. This is widely used to provide visual feedback, such as changing the color of a button or link when the user hovers over it.

The `lang` pseudo-class

targets elements that have a specific language code in their lang attribute. This enables you to create language-specific styling rules, ensuring that content looks appropriate and consistent across different languages.

```
p:lang(fr) { font-style: italic; }


<p lang="fr">
   Adieu
</p>
<p lang="jw">
   Sugeng rawuh
</p>
```

The **negation** pseudo-class

takes a simple selector as an argument and applies styles to all elements that do not match that selector. It’s especially useful when you want to apply a style broadly but need to exclude certain elements from that rule.

`:not(p) { }` – all elements except paragraphs **tags**

`:not(.intro) { }` – all elements except those **with class** .intro

`:not(#news) { }` – all elements except those **with id** #news

`:not(:lang(fr)) { }` – all elements **except** those with the French **language**

`:not([disabled]) { }` – all elements except those **without** the disabled **attribute**

`p:not(.intro) { }` – all paragraphs **except** those with the class `.intro`

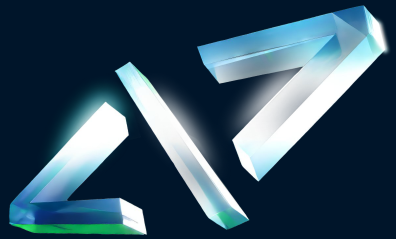
Structural pseudo-classes

are selectors that allow you to apply styles to elements based on their position within the parent element or the document structure as a whole. These pseudo-classes are particularly useful for targeting elements in complex layouts where the order or structure of elements matters.

```
:first-child { }
:only-child { }
:nth-child(3n) { }
```

! If the document structure changes, the structural pseudo-class might apply to a different element or potentially to no element at all.

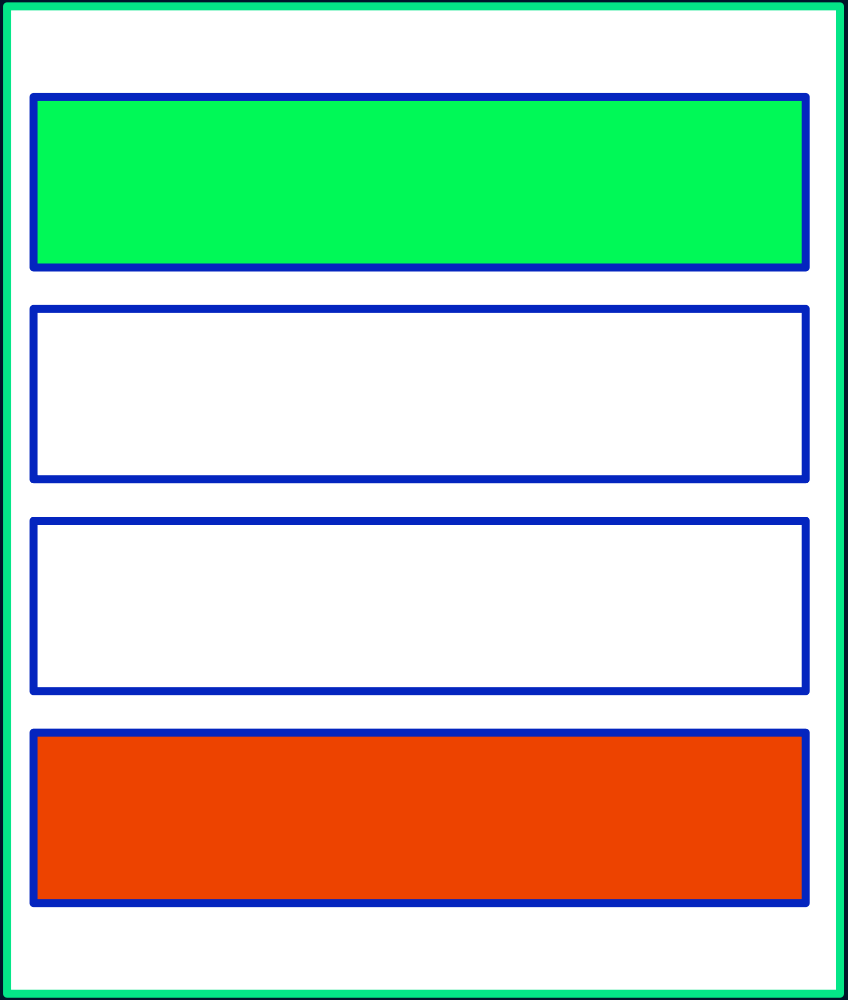
It can sometimes be difficult to determine exactly which element the styles will be applied to.

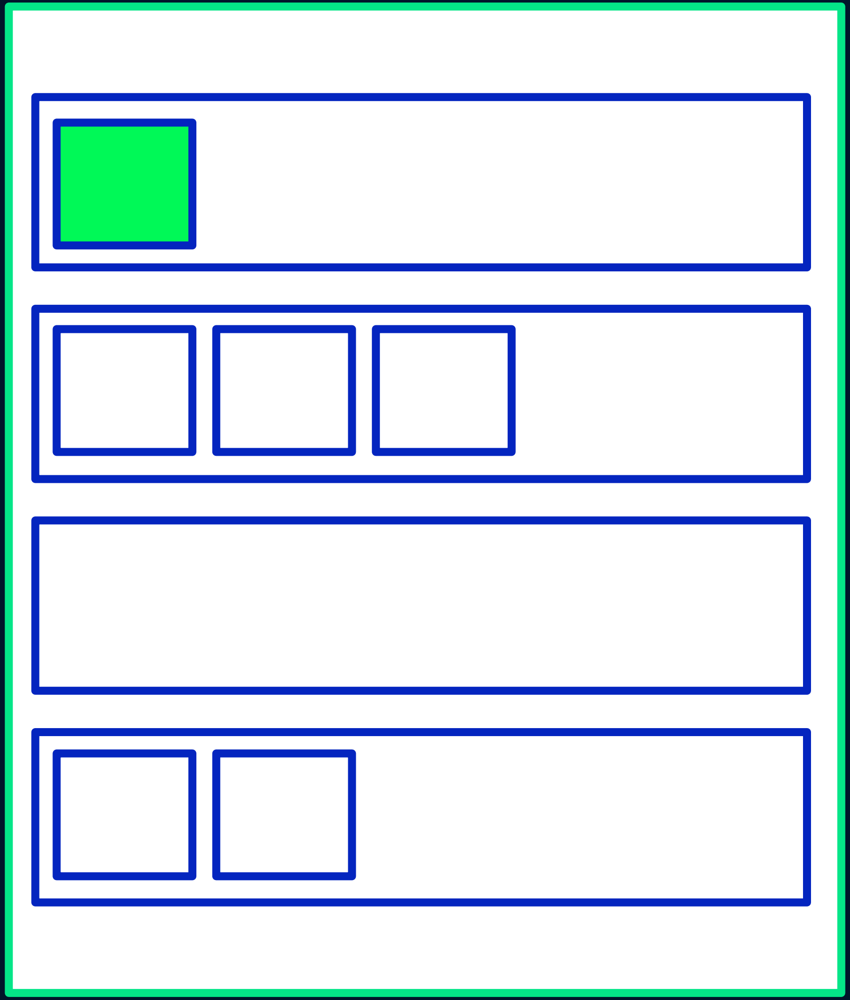


:first-child
:last-child

– Selects the element that is the first/last child of another element.

```
article:first-child { }
article:last-child { }

<section>
   <article> 1 </article>
   <article> 2 </article>
   <article> 3 </article>
   <article> 4 </article>
</section>
```



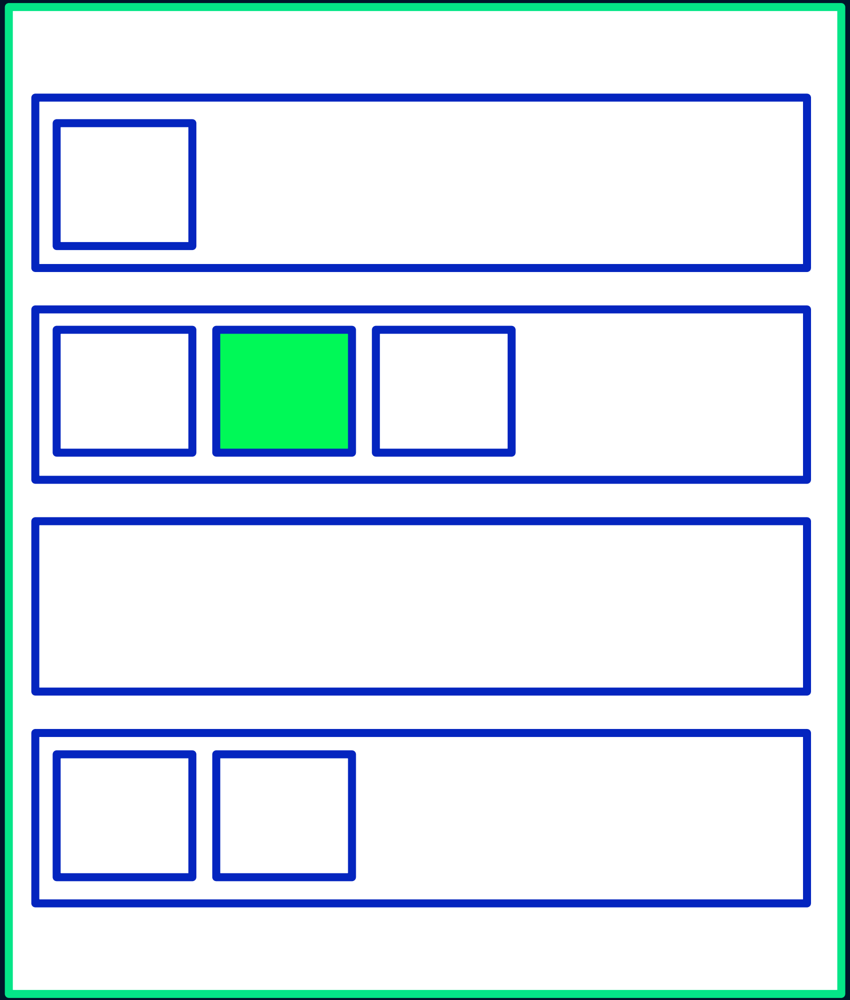


pseudo-class

:only-child

– Selects an element that is the only child of another element.

```
div:only-child { }

<article>
   <div> 1 </div>
</article>
```



pseudo-class

:only-of-type

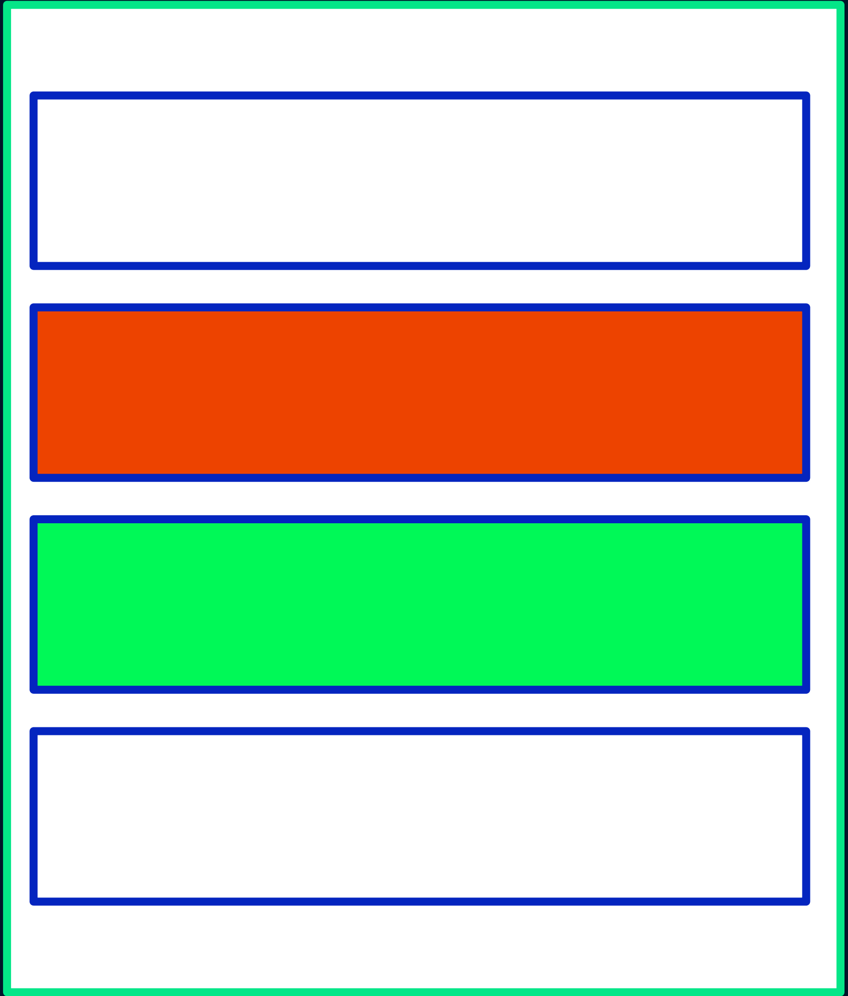
– Selects an element that is the **only** element of its **type** within its parent.

```
p:only-of-type { }

<article>
   <div> 1 </div>
   <p> 1 </p>
   <div> 1 </div>
</article>
```

:first-of-type
:last-of-type

– Selects an element that is the first/last of its type within its parent element.

```
p:first-of-type { }
p:last-of-type { }

<section>
   <article> 1 </article>
   <p> 2 </p>
   <p> 3 </p>
   <article> 4 </article>
</section>
```

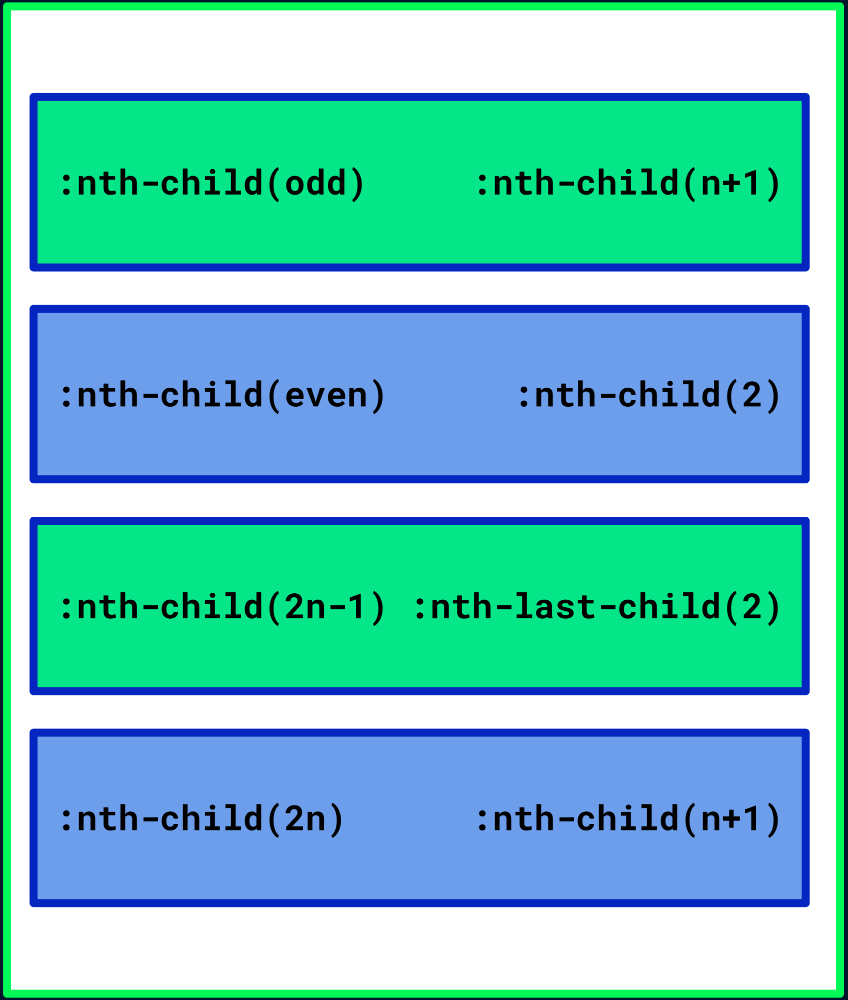


:nth-child(n)
:nth-last-child(n)

– Selects **specific child elements** in a parent element starting from the beginning or the end.

n:

- **number**
- **number + n** (selects every **n-th** element)
- expression with **+/-** (allows starting from an element other than the first)
- **even** (all even elements)
- **odd** (all odd elements)

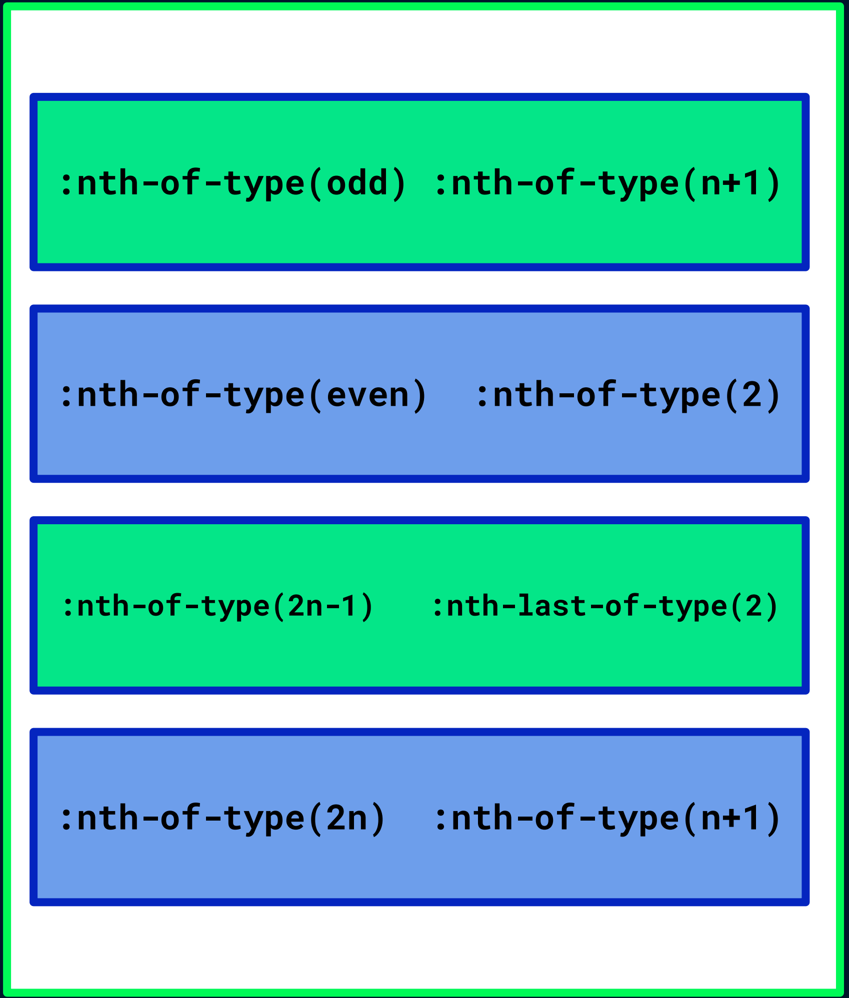


:nth-of-type(n)
:nth-last-of-type(n)

– Selects elements of a **specific type in the parent** element starting from the beginning or the end.

n:

- **number**
- **number + n** (selects every **n-th** element)
- expression with **+/-** (allows starting from an element other than the first)
- **even** (all even elements)
- **odd** (all odd elements)



:root

– Selects the root element of the document (тег `<html>`).

:empty

– Selects an element that has no content or child elements (an empty element).

A space is already a character, so the tag is no longer considered empty.
It also applies to input elements where no value has been entered.

```
:root { }

<html>
    <head> 1 </head>
    <body> 1 </body>
</html>


p:empty { }

<article>
    <p> 1 </p>
    <p> </p>
    <p></p>
    <p><span></span></p>
</article>
```

PSEUDO-ELEMENTS

Pseudo-elements

– **(fake elements)** Allow styling elements that are not in the document tree.

`::-webkit-scrollbar` – styles the scrollbar

\+ Other pseudo-elements of the form `::-webkit-scrollbar-*`, are used only with prefixes and only in `webkit` browsers

```
.invisible-scrollbar::-webkit-scrollbar {
  display: none;
}
```

Pseudo-elements for text

`::first-line` – styles the first line of text

`::first-letter` – styles the first letter of text

```
p::first-line { }
p:first-letter { }
<p>
   This is the first line
   of a paragraph of text
</p>
```

Pseudo-elements that Create a New Element

`::before` – A new element is created at the beginning of the element, before the content

`::after` – A new element is created at the end of the element, after the content

- An element can have only one `::before` and one `::after`
- These pseudo-elements are inserted into the document flow and occupy space
- They are visible in the inspector but cannot be accessed via JavaScript
- Any CSS properties can be applied to them.
- They can only be applied to elements with closing tags
- Often used to add decorative elements

content

– replaces content with a generated value.

This is a **required** property for the **before** and **after** pseudo-elements

```
p::before {
  content: “”;
}
```

Text	`“hello”;`
Image	`url(pic.png);`
Attribute (displays the value of an attribute as text)	`attr(cite);`
Counter	`counter(list-order);`
Nothing	`"  ";`
Special characters	`"\21E6";`
Emoji	`"🎮";`
Gradient	`linear-gradient(#e66465, #9198e5);`

content

In the **content** property, you can combine different values:

```
p:before {
  content: "class: " attr(class);
}
```

The **text "class: "** will be placed before the paragraph content, followed by the list of **all class attribute** values for that paragraph.

Cannot contain HTML tags:

```
p:before {
  content: <p>test<p>;
}
```

Pseudo-elements for Lists

- Usage of counters in lists
- Styling list markers

`::marker` – Styling list markers.

```
ol {
  counter-reset: section;
}
li::before {
  counter-increment: section;
  content: counter(section);
}
li::marker { }
```